Abu Jehad Arafat

PhD Applicant · Robust Perception and State Estimation for Autonomous Systems · M.Sc. Data Science & ML

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RESEARCH INTERESTS

Seventeen years of operating in safety-critical aviation, including flying and decision-making in degraded visibility, motivate my interest in machine perception and state estimation that stay reliable when their inputs do not.

- Robust perception in degraded visual environments: per-image difficulty estimation, detector failure modes and severity-aware adaptive inference. M.Sc. thesis.
- Full-stack autonomy for aerial systems: extending perception toward planning and control for autonomous and unmanned aircraft.

EDUCATION

M.Sc. in Data Science & Machine Learning

State University of Bangladesh · Degree awarded September 2026 · Thesis 7 credits

Coursework: Machine Learning for Data Science, Artificial Intelligence for Data Science, Big Data Analysis, Multivariate Analysis, Advanced Sampling Methods & Statistical Inference, Applied Regression & Time-Series Forecasting, Linear Algebra & Statistics, Database Management Systems, Programming for Data Science.

Aviation Instructors' Postgraduate Diploma

Bangladesh University of Professionals · Instructional technique, air law, applied aerodynamics, human performance and limitations, navigation

B.Sc. in Aeronautics

Bangladesh University of Professionals · Aerodynamics, aero-engines, meteorology, aviation physiology, airmanship

M.SC. THESIS RESEARCH

Dark-Channel Fog Severity as a Safety Signal: Failure Modes and Limits on Real Fog

M.Sc. thesis, State University of Bangladesh · Supervisor: Prof. Pintu Chandra Shil

- Audits the dark-channel fog-severity score as a training-free, per-image predictor of object-detection difficulty: calibrated on 2,000 synthetic-fog images, tested on real fog (RTTS, 4,245 images) against three clear-trained detectors (YOLO-nano, YOLOv8-S, RT-DETR-L), with no dehazing or enhancement stage.
- Headline finding: where the score reports near-clear, 39.9% of images are still detection failures, and the best of 11 appearance descriptors separates them at only AUC 0.686. That is a false-reassurance failure mode shared by every free estimator tested. Practical rule: trust a high severity score, never a low one.
- For a robot or autonomous aircraft, a difficulty signal that is trustworthy only in one direction changes how perception uncertainty should be used by planning and control.

PUBLICATIONS

- A. J. Arafat and P. C. Shil, "Dark-Channel Fog Severity as a Safety Signal: Failure Modes and Limits on Real Fog," submitted to IEEE Transactions on Intelligent Vehicles, 2026 (under review). Code and figures: github.com/aaaraafaat/perception-difficulty-dve

RESEARCH & DEVELOPMENT PROJECTS

Digital Flight Console (PC-based aviation training device, PT-6) · Team leader, BAF R&D

Project lead for a PC-based training device (Core i9 / RTX 4090 workstation, triple QHD displays, IR head and eye tracking, VPC stick, throttle and pedals) on Prepar3D/MSFS with a custom fixed-wing model. Researched, selected and purchased all the COTS hardware, then designed the full cockpit layout, including the controls arrangement. Presented the design for approval. Managed the budget and directed the build team through to a fielded system, at about 1/96th the cost of a full mission simulator.

BAF Mission Assistant (automated mission-map preparation) · Team member (UI design), BAF R&D

Secured mission-planning platform on a continuously updated global map, producing an operational map with standardized navigation data in under 15 minutes across aircraft types.

TEACHING & INSTRUCTION

- **Flying Instructor, Flying Instructors' School and Bangladesh Air Force Academy:** 566 instructional flight hours. In-flight and ground instruction to 65 cadet pilots and 26 instructor pilots. Designed and conducted assessments, check-rides and technical briefings, and lectured Air Law and Instructional Technique to the 64th Flying Instructors' Course.

PROFESSIONAL EXPERIENCE (SELECTED)

Project Officer, Air Warfare Centre Project

Directorate of Cyberwarfare & Information Technology, Air Headquarters, Bangladesh Air Force (attached to the Air Warfare Centre) · Dhaka, Bangladesh

- Manage the Air Warfare Centre's USD 10M simulation, mission-planning, flight-debrief and wargaming integration program.

Staff Instructor / Flight Commander, Operations & Training Program Manager

Flying Instructors' School, Bangladesh Air Force · Bogura, Bangladesh

- Planned and ran the daily flying-training program of a training squadron (65 trainee pilots and 26 instructors in one year), allocating aircraft, instructors and sorties against a tight syllabus for ab-initio and instructor-pilot courses.

Earlier service: Staff Officer, Aviation Safety with the UN Hybrid Operation in Darfur (UNAMID), Sudan. Ran the mission aviation-safety program in a conflict zone and trained 500+ personnel. Also Officer Commanding Flight & Ground Safety Squadron at a forward operating base. Operational fighter pilot and Base Adjutant for 400+ personnel.

TECHNICAL SKILLS

- **Programming & ML:** Python (pandas, scikit-learn, OpenCV) and PyTorch at M.Sc.-thesis level (YOLO, RT-DETR), experiment design and evaluation, LaTeX, Zotero, Git, Colab.
- **Tools:** Builds research and training tools with AI-assisted development (Claude Code). Git, GitHub, HTML/JS.

HONOURS, QUALIFICATIONS & LANGUAGES

- **Honours and qualifications:** Chief of Air Staff Commendation, Bangladesh Air Force · United Nations Medal (UNAMID), for peacekeeping service · Best in Flying trophy, BAF Academy Basic Flying Training · Qualified Flying Instructor · Flight Safety Officers' Course · Aviation Risk Management (UN) · IBM Data Science Professional.
- **Languages:** English, CEFR C1 (TOEFL iBT 103/120) · Bangla, native. Former military pilot and flying instructor on ten aircraft types.

Referees available on request.